Renan Alves de Oliveira

AI & MLOps Engineer | Satellite Imagery & Geospatial ML in Production | Ph.D. in Astrophysics

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Professional Summary

I enjoy solving complex problems, whether analyzing data from the cosmos or simply coding a solution to a problem. My strong foundation in physics and computational modeling has enabled me to transform raw data into value. My expertise ranges from developing robust ETL/ELT data pipelines to deploying AI models for specific domains: astrophysics, fitness metrics, LLMs, and geospatial analysis.

Work Experience

AI Developer Specialist — **March 2026 – Present**
BemAgro S.A. — *Ribeirão Preto, SP*

- Develop and support an end-to-end production Python pipeline for geospatial analysis using artificial intelligence.
- Improved operational reliability and troubleshooting with structured per-step status, timing and peak-memory metrics, persistent logs, distributed tracing, stable exit codes, and structured error handling for production failures.
- Provision and operate compute across cloud (OCI, Azure, GCP).

AI Specialist — **May 2025 – January 2026**
Turing — *Campinas, SP*

- Deployed a scalable multi-tenant API on a Kubernetes cluster on AWS, serving multiple client workloads from a single service.
- Deployed to production multi-agent systems on AWS managing 200+ specialized agents and 1K+ API integrations (LangChain, LlamaIndex, n8n, WSO2) with RAG, web search, and third-party services to automate complex back-office and service optimization workflows at scale.
- Created and deployed APIs for Robotic Process Automation for GUI apps and LLMs.

AI Specialist — **January 2025 – January 2026**
Psyche Aerospace — *Campinas, SP*

- Developed computer vision pipeline processing multispectral drone and satellite imagery to compute vegetation indices (NDVI, SAVI, EVI, among others) for automated crop health assessment.
- Built AWS infrastructure (EC2, SageMaker, Bedrock, EKS) to automate web scraping and fine-tuning of Microsoft Phi-4 on large-scale agricultural corpus, creating domain-specific LLM for agribusiness.
- Implemented high-performance RAG system using GPU-accelerated FAISS (FLAT index) over 3 million+ documents, achieving 95-100% recall with low latency for hallucination detection and real-time information retrieval in production.

Project Leader — **September 2023 – December 2024**
Samsung Electronics — *Campinas, SP*

- Led end-to-end development of advanced fitness metrics for Samsung Galaxy Watch (VO2max/%VO2max, AT/AnT, HRmax estimations), coordinating a 12-person cross-functional team to deliver features to millions of active users.
- Coordinated sensor data integration (HR, accelerometer, GPS) and algorithm validation with research teams, maintaining technical documentation and achieving on-time delivery rate across product releases.

Data Engineer Sr — **August 2022 – December 2024**
Samsung Electronics — *Campinas, SP*

- Built automated ETL pipelines (using Pandas, PostgreSQL) for processing time-series health metrics (HR, speed, VO2/VCO2), improving model training efficiency and reducing manual preprocessing time.
- Implemented data quality framework with automated validation and error-handling in CI/CD (GitHub Actions), reducing production data errors and ensuring consistent data integrity across environments.
- Redesigned data ingestion and storage architecture (MinIO S3-compatible object storage, PostgreSQL) with engineering team, improving data retrieval speed and enabling real-time time-series monitoring dashboards in Grafana and Power BI.

Researcher — **August 2018 – April 2024**
Universidade Federal do Espírito Santo — *Vitória, ES*

- Administered the research group's HPC cluster on Enterprise Linux (CentOS), ensuring uptime and resource allocation for critical scientific computations.
- Developed and deployed an interactive dashboard (Streamlit, later React/TypeScript) backed by a structured database for astronomical data visualization, served from a remote server for distributed collaborators.

- Mentored students in Python, Unix, and Shell, and optimized computational workflows with Cython, Numba, and parallelization techniques.

Researcher Analyst **November 2019 – May 2020**
Simons Foundation (Flatiron Institute) *New York, NY*

- Developed a high-performance neural emulator (PyTorch, deployed via ONNX to OpenVINO) to accelerate dark matter simulations, enabling single-GPU predictions of particle evolution using a StyleGAN-based architecture for parameter conditioning. Validated the model's statistical accuracy (correlation and transfer functions) against semi-analytical and standard methods. Presented accepted paper at NeurIPS 2020 Machine Learning and the Physical Sciences workshop.

Researcher **August 2016 – July 2018**
Universidade Estadual de Londrina *Londrina, PR*

- Validated non-Gaussianity and isotropy assumptions in cosmological models through statistical analysis (Pearson's χ^2 test) of Planck satellite Cosmic Microwave Background data. To facilitate this work, developed polyMV, a Python/C software library for efficient conversion of spherical harmonic coefficients to Cartesian vector representations, with applications beyond cosmology.

Junior Researcher **May 2014 – July 2014**
Hydrologic Research Center *Del Mar, CA*

- Modeled critical failure conditions for landslide initiation through geotechnical slope stability analysis, applying continuum mechanics and numerical methods as part of a Science Without Borders internship.

Education

Doctor in Astrophysics, Cosmology and Gravitation **August 2018 – April 2024**
Universidade Federal do Espírito Santo *Vitória, ES*

- Thesis: *Probing cosmology with an eye on Rubin: from strong lensing to the large scale structure of the universe*

Master in Physics **August 2016 – July 2018**
Universidade Estadual de Londrina *Londrina, PR*

- Dissertation: *Testes de isotropia utilizando vetores de multipolo em pequenas escalas*

Bachelor of Science in Applied Physics **July 2013 – May 2014**
California State University, San Marcos *San Marcos, CA*

- *Science Without Borders Program*

Bachelor in Physics **March 2010 – March 2016**
Universidade Estadual de Londrina *Londrina, PR*

Technical Skills

DevOps & Cloud: AWS, Azure, CI/CD Pipelines, Docker, GitHub Actions, Google Cloud Platform, Infrastructure as Code, Kubernetes, Linux (CentOS/RHEL, Ubuntu), Oracle Cloud Infrastructure, Terraform
Programming & Testing: C, Cython, Numba, Python, pytest, R, Shell, TypeScript
Operations & Observability: Configuration Management, Distributed Systems, Distributed Tracing, Grafana, Incident Response, Logging, Monitoring, Performance Tuning, Troubleshooting
Data Engineering: Data Pipelines, DVC, ETL, MinIO (S3-compatible object storage), MLflow, PostgreSQL, STAC
AI & Machine Learning: Agents, LLM Fine-tuning, Multimodal Prompting, ONNX, PyTorch, RAG
Satellite & Space Data: Earth Observation (Sentinel-2, Landsat, Planet), Space Telescopes (Hubble, JWST, Planck), Multispectral Imagery, Remote Sensing, Geospatial Analysis
Ways of Working: Agile (Kanban, Scrum/Sprints), Code Review, Confluence, Git, GitHub, GitLab, Jira, Mentoring, Technical Documentation
Data Analysis: Bayesian Inference, Meta-Analysis, Monte Carlo Simulation, Statistical Testing
Visualization & Reporting: DataViz, Power BI, React, Scientific Writing, Streamlit
HPC & Optimization: Code Optimization, High-Performance Computing, Parallel Computing

Selected Publications

- **Alves de Oliveira, R.** et al. "The Last Stand Before Rubin: consolidated sample of strong lensing systems". *arXiv preprint*, 2025. arXiv:2509.09798
- **Alves de Oliveira, R.** et al. "Field-level neural network emulator for cosmological n-body simulations". *The Astrophysical Journal*, 2023. 10.3847/1538-4357/acdb6c
- **Alves de Oliveira, R.** et al. "CMB statistical isotropy confirmation at all scales using multipole vectors". *Physics of the Dark Universe*, 2020. doi:10.1016/j.dark.2020.100608

Languages & Certifications

Languages: Portuguese (Native), English (Full Professional Proficiency), Spanish (Comprehension)
Certifications: Data Science (Unicamp)